

SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1

Research Paper Review / Literature Survey

**“Intelligent AI-Based Airport Baggage Tracking
and Management System”**

A Literature Survey Report

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Declaration of Academic Integrity

I, RR Thiyagesh (Reg. No. 192572007), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Abstract

Airports worldwide handle billions of pieces of checked baggage every year, and even a small percentage of mishandling translates into substantial passenger inconvenience and financial loss for airlines. Conventional baggage-handling infrastructure, built primarily around barcode scanning and periodic manual checkpoints, offers limited visibility into the real-time location of a bag and almost no ability to anticipate delays before they occur. This literature survey examines the body of research on RFID-, IoT-, blockchain-, and vision-based baggage tracking systems to understand how the field has progressed from simple identification tagging toward predictive, AI-driven logistics management. Eight primary studies spanning 2008 to 2025 are reviewed and compared on methodology, enabling technology, dataset or tooling, and reported outcomes. The comparative analysis shows that while real-time location tracking is now a mature capability, very few existing systems incorporate genuine predictive analytics for delay forecasting, automated rerouting, or proactive passenger communication. Building on these findings, this report proposes an architecture for an Intelligent AI-Based Airport Baggage Tracking and Management System that layers a machine-learning analytics engine over an RFID/IoT sensing network and a cloud data store, and it outlines the research gaps and future directions that motivate this design.

1. Introduction

1.1 Industry Context

Airports manage millions of passenger bags each year, making efficient baggage handling crucial for operational performance and customer satisfaction. Radio-Frequency Identification (RFID), the Internet of Things (IoT), Artificial Intelligence (AI), and cloud computing have collectively enabled a shift toward real-time baggage tracking, automated sorting, and predictive logistics management. Global aviation bodies have pushed the industry toward smart-tagging standards precisely because manual, barcode-based handling cannot keep pace with rising passenger volumes and increasingly complex hub-and-spoke routing networks.

1.2 Problem Statement

Baggage delays, loss, and misrouting create inconvenience for passengers and increase operational costs for airlines. Traditional baggage-handling systems provide limited real-time tracking and virtually no predictive capability. An AI-powered baggage management platform is therefore needed that can monitor baggage movement using RFID, optimize baggage routing, predict delays using AI, notify passengers automatically, reduce baggage misrouting, and generate operational analytics — improving handling efficiency while reducing passenger complaints.

1.3 Objectives

- Track baggage in real time across every checkpoint of the airport journey.
- Optimize baggage routing between check-in, sorting, and aircraft loading.
- Predict baggage delays using AI models trained on checkpoint and flight data.
- Notify passengers automatically about the status and location of their baggage.
- Reduce baggage misrouting through automated, sensor-verified handoffs.
- Generate operational reports and analytics for airport and airline staff.
- Improve overall airport logistics and passenger satisfaction.

1.4 Expected Outcomes

- Reduced baggage loss and faster baggage delivery.
- Improved airport operations and better passenger experience.
- Increased baggage-tracking accuracy and lower operational costs.
- Enhanced logistics efficiency and improved airline service quality.

2. Literature Survey

This section reviews eight representative studies on airport baggage tracking, selected to span the major technological approaches reported in the literature: RFID-only identification, RFID combined with cloud or GSM/GPS connectivity, blockchain-secured tracking, machine-vision-based anomaly detection, and IoT-enabled robotic monitoring.

2.1 DeVries (2008) – The State of RFID for Effective Baggage Tracking in the Airline Industry

International Journal of Mobile Communications, Vol. 6, No. 2

Objective: to examine the feasibility and business case for deploying RFID in place of barcode tags across the airline baggage-handling value chain.

Methodology / Technology: a qualitative industry analysis of early RFID trials, comparing passive-tag read accuracy against optical barcode scanning and evaluating the cross-organizational cooperation required between airports and airlines to fund shared RFID infrastructure.

Key Findings: RFID substantially improves first-read accuracy over barcodes, but adoption is slowed less by the technology itself than by the difficulty of building a shared business case across multiple airport stakeholders who each bear part of the infrastructure cost.

2.2 Singh, Meshram, Gujar & Wankhede (2016) – Baggage Tracing and Handling System Using RFID and IoT for Airports

IEEE Int. Conf. on Computing, Analytics and Security Trends (CAST), pp. 466–470

Objective: to design a working prototype that couples RFID identification with an IoT cloud backend so that a bag's real-time position can be queried by its owner.

Methodology / Technology: a two-location hardware prototype (check-in and check-out) using RFID readers, a secure tag-generation algorithm, and a cloud server that stores each scan event against a unique baggage ID retrievable through a mobile interface.

Key Findings: Step-wise RFID scanning at checkpoints, combined with cloud storage, reduced loss incidents in the prototype and enabled passengers to retrieve baggage status using a unique ID from any location, though the system offered no forward-looking delay prediction.

2.3 Si et al. (2021) – BEI-TAB: Enabling Secure and Distributed Airport Baggage Tracking with Hybrid Blockchain-Edge System

Wireless Communications and Mobile Computing, Vol. 2021, Article ID 2741435

Objective: to overcome the data-leakage and single-point-of-failure risks of centralized cloud storage by combining RFID with a consortium blockchain and edge computing.

Methodology / Technology: an RFID-plus-blockchain architecture in which every checkpoint scan is written to a distributed ledger via edge nodes, physically isolating baggage data and removing dependence on one central database.

Key Findings: The hybrid blockchain-edge design improved data confidentiality and transmission security relative to centralized-cloud approaches and reduced labour cost by automating the recording step, though at the cost of higher computational overhead at each edge node.

2.4 Chen et al. (2024) – Research on Airport Baggage Anomaly Retention Detection Technology Based on Machine Vision, Edge Computing, and Blockchain

IET Blockchain

Objective: to detect abnormal retention (bags stuck or delayed at a stage) using camera-based machine vision rather than relying purely on RFID scan gaps.

Methodology / Technology: a CNN-based multi-camera vision pipeline for tracking baggage and passengers across cameras, integrated with edge computing for low-latency inference and blockchain for tamper-resistant logging of anomaly events.

Key Findings: Machine-vision anomaly detection identified stuck or misrouted baggage earlier than scan-gap heuristics alone, and combining it with edge inference kept detection latency low enough for operational use, though the approach requires denser camera coverage than RFID-only systems.

2.5 Thiyaagu, Sarath Chandra, Nandan & Manesh Vardhan Reddy (2018) – IoT Based Airport Baggage System

Journal of Emerging Technologies and Innovative Research (JETIR), Vol. 5, Issue 10

Objective: to propose a low-cost architecture for automating baggage handling that focuses specifically on RFID-based tracking of passenger baggage through the terminal.

Methodology / Technology: an RFID reader network integrated with an IoT communication layer, describing tag structure, reader placement, and the destination-airport handling process end to end.

Key Findings: The proposed IoT-RFID architecture demonstrated that even low-cost reader hardware can provide checkpoint-level visibility, but the study stopped short of addressing predictive analytics or passenger-facing notification.

2.6 Srinivasa Rao, Madhava Reddy, Vijaya Bhaskar & Sreenath (2023) – Baggage Tracking and Handling by RFID and IoT in Airports Using NodeMCU

Mathematical Statistician and Engineering Applications

Objective: to build an inexpensive, microcontroller-based (NodeMCU) baggage tracking prototype suitable for smaller or resource-constrained airports.

Methodology / Technology: a NodeMCU-driven RFID reader network installed at departure and destination sections, where each conveyor section reads the tag and forwards status to a cloud dashboard for passenger monitoring.

Key Findings: The NodeMCU-based design showed that section-wise RFID scanning on the conveyor path can be implemented at low hardware cost while still giving passengers cloud-based visibility into baggage status, making it attractive for budget-constrained deployments.

2.7 Systematic Literature Review (2021) – Adoption of Industry 4.0 Technologies in Airports

arXiv:2112.14333

Objective: to systematically map how Industry 4.0 technologies, including RFID, WSN, IoT, and predictive maintenance analytics, have been adopted across airport logistics functions.

Methodology / Technology: a systematic review methodology synthesizing published case studies, including RFID-tagged cargo monitoring, IoT-enabled luggage localization tags, and a Heathrow Airport pilot that used IoT sensors and data analytics for predictive maintenance of baggage-handling conveyor carts.

Key Findings: The review found that RFID and IoT are well established for baggage identification and cargo monitoring, but predictive-analytics applications remain concentrated on equipment maintenance rather than on forecasting baggage delay or misrouting risk directly — an application-level gap this survey also identifies.

2.8 IoT-Enabled Smart Robotic Baggage Monitoring and Tracking System (2025)

IAES International Journal of Artificial Intelligence

Objective: to explore how IoT-enabled robotic and AI components can extend baggage monitoring beyond static RFID checkpoints toward proactive, self-adjusting logistics.

Methodology / Technology: an IoT framework combining RFID/barcode identification with cloud-based usage-pattern learning, positioned as a step toward machine-learning-driven prediction of traveller baggage needs and automated handling decisions.

Key Findings: The study argues that AI and machine learning can proactively address baggage-handling concerns and improve the passenger journey, but notes that most deployed systems still rely on reactive scan-based tracking rather than trained predictive models, reinforcing the case for the AI analytics layer proposed in this report.

3. Proposed System Architecture

Drawing on the strengths and gaps identified across the reviewed literature — particularly the consistent absence of predictive analytics noted in Sections 2.5, 2.7, and 2.8 — this report proposes a four-layer architecture for the Intelligent AI-Based Airport Baggage Tracking and Management System. The sensing layer captures RFID, IoT, and vision data at each physical checkpoint; the edge layer performs local filtering and timestamping to reduce cloud load; the cloud/intelligence layer stores checkpoint events (optionally on a distributed ledger, following the security rationale of Si et al. [3]) and runs an AI/ML engine that predicts delay and misrouting risk; and the application layer surfaces this intelligence to passengers and airport operations staff.

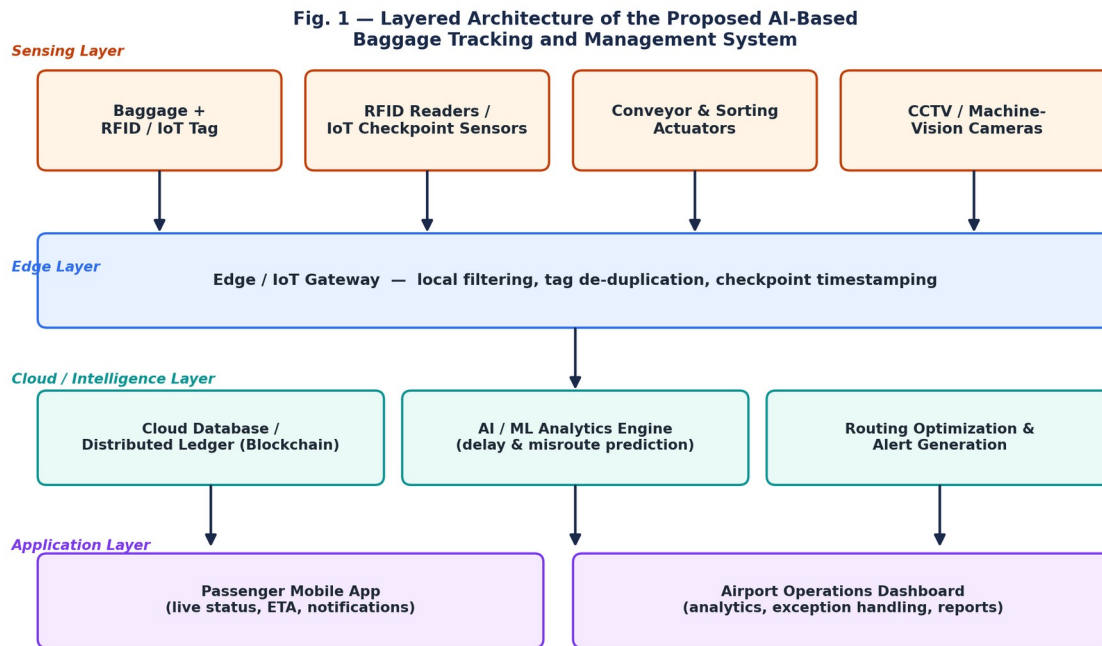


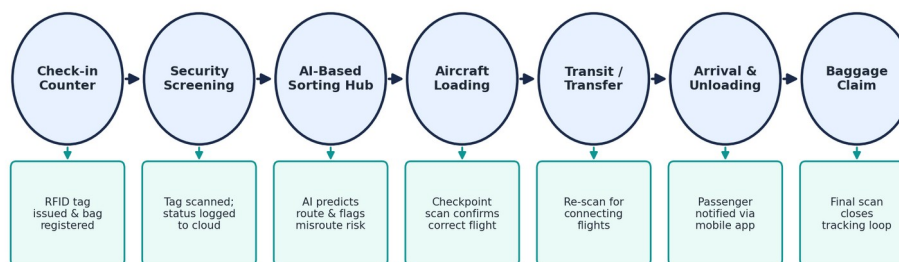
Figure 1. Layered architecture of the proposed AI-based baggage tracking and management system.

Unlike the RFID-plus-cloud designs of Singh et al. [2] and Srinivasa Rao et al. [6], the addition of a dedicated AI/ML analytics engine allows the system to move from reporting where a bag is to forecasting where it is likely to end up and whether it is at risk of missing its connection — directly addressing Objective 3 (predict baggage delays using AI) stated in Section 1.3.

4. Baggage Journey and Tracking Workflow

Figure 2 traces a bag through the seven operational stages identified across the reviewed studies — check-in, security screening, sorting, aircraft loading, transit/transfer, arrival, and claim — and shows the checkpoint event that is captured at each stage. This workflow view mirrors the two-location (check-in/check-out) prototypes of Singh et al. [2] and the section-wise conveyor scanning of Srinivasa Rao et al. [6], but adds an AI-based sorting decision at the third checkpoint and a passenger-notification trigger at arrival, neither of which appears as a built-in feature in the systems surveyed in Section 2.

Fig. 2 — End-to-End Baggage Journey with AI-Enabled Tracking Checkpoints



Each checkpoint writes a timestamped event to the cloud/blockchain layer; the AI engine continuously re-evaluates predicted arrival time and misrouting risk after every scan.

Figure 2. End-to-end baggage journey with AI-enabled tracking checkpoints.

5. Comparative Literature Survey Table

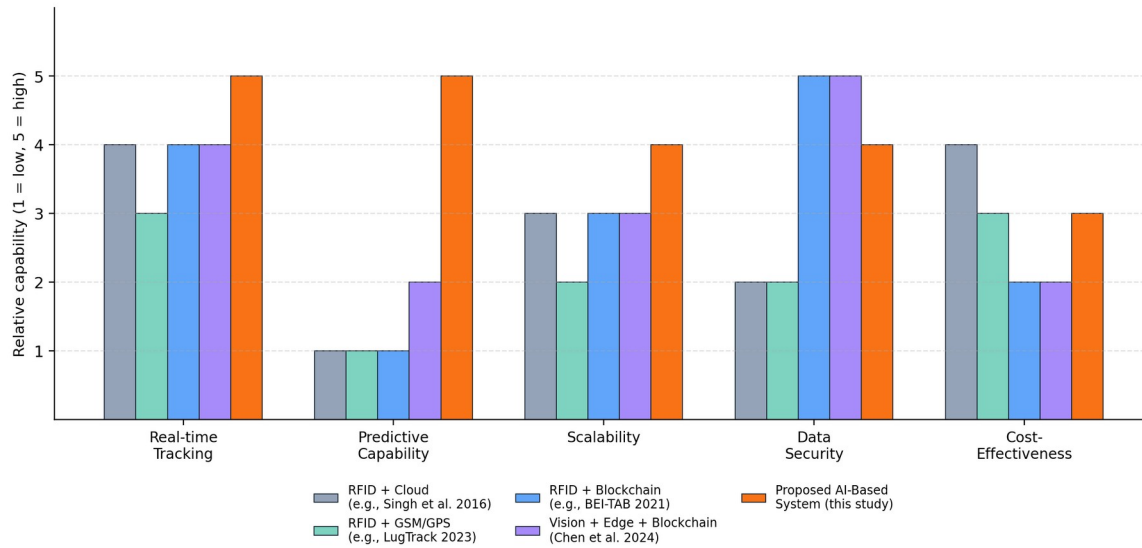
Table 1 consolidates the eight reviewed studies against seven comparison parameters — author/year, research problem, methodology, dataset/tools, key findings, limitations, and the resulting research gap — as required for a structured literature survey.

Table 1. Comparative summary of reviewed literature.

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
DeVries (2008)	RFID vs. barcodes for baggage ID	Qualitative RFID trial analysis	Industry trial reports	RFID improves read accuracy over barcodes	No prototype; cross-org cost sharing	No predictive/AI component
Singh et al. (2016)	Real-time bag location visibility	RFID+IoT cloud prototype, 2-location test	RFID readers, cloud server	Reduced loss; self-lookup by ID	Only 2 checkpoints scanned	No delay prediction/auto-notify
Si et al. (2021)	Data security in centralized tracking	RFID+consortium blockchain+edge	Blockchain ledger, edge nodes	Improved confidentiality vs. cloud	Higher edge compute overhead	No AI routing/forecasting layer
Chen et al. (2024)	Detecting stuck/delayed baggage	CNN vision + edge + blockchain	CCTV, CNN, edge inference	Earlier anomaly detection than scan gaps	Needs dense camera coverage	Vision data not fused with RFID
Thiyagu et al. (2018)	Low-cost terminal RFID tracking	RFID reader network + IoT	RFID tags/readers	Checkpoint visibility at low cost	No predictive analytics/app	No passenger notification
Srinivasa Rao et al. (2023)	Budget-constrained tracking	NodeMCU RFID on conveyor	NodeMCU, RFID, dashboard	Low-cost section-wise visibility	Limited to conveyor path	No AI/ML misroute prediction
Systematic Review (2021)	Mapping Industry 4.0 adoption	Systematic literature synthesis	Case studies incl. Heathrow	Predictive analytics used for maintenance mostly	Review-level; no system built	Predictive analytics rarely on baggage flow
Robotic IoT Study (2025)	Reactive to proactive tracking	IoT + usage-pattern learning	IoT sensors, cloud platform	Argues AI/ML can proactively flag issues	Most deployments still reactive	Few systems combine ML with live data

6. Comparative Capability Analysis

To visualize how the reviewed approaches trade off against one another, Figure 3 scores five representative system types — RFID+Cloud, RFID+GSM/GPS, RFID+Blockchain, Vision+Edge+Blockchain, and the proposed AI-based system — on five parameters drawn from the rubric for comparative analysis: real-time tracking, predictive capability, scalability, data security, and cost-effectiveness. The scores are qualitative, synthesized from the reported strengths and limitations in Sections 2 and 5 rather than measured on a common benchmark, since the underlying studies do not share a uniform test environment.

Fig. 3 — Qualitative Comparison of Approaches from the Reviewed Literature Across Key Evaluation Parameters*Figure 3. Qualitative comparison of reviewed approaches across key evaluation parameters.*

The comparison highlights a consistent pattern: every RFID- and vision-based approach reviewed scores high on real-time tracking but low on predictive capability, while data security improves markedly once blockchain is introduced (Si et al. [3]; Chen et al. [4]) at some cost to real-time performance and system cost. The proposed AI-based system is designed to retain strong real-time tracking and data security while raising predictive capability substantially by adding a trained analytics layer — the parameter most consistently identified as missing across the literature.

7. Critical Analysis and Research Gaps

7.1 Limitations of Existing Approaches

- Reactive rather than predictive: nearly all reviewed systems [1], [2], [5], [6] report a bag's location after the fact but do not forecast delay or misrouting risk before it happens.
- Fragmented data sources: RFID/IoT tracking [2], [5], [6] and machine-vision anomaly detection [4] are studied separately; no reviewed system fuses both streams into a single model.
- Security-performance trade-off: blockchain-based designs [3], [4] improve data integrity but introduce computational overhead that has not been evaluated at full-scale airport throughput.
- Limited passenger-facing intelligence: most systems expose only a status lookup by ID [2], [6] rather than proactive notifications or predicted arrival times.
- Narrow evaluation environments: several studies are single-location prototypes or two-checkpoint testbeds [2], [5], [6], so scalability claims are not verified against real multi-terminal, multi-airline traffic.

7.2 Identified Research Gap

Across the literature reviewed in Section 2 and summarized in Table 1, the recurring and largest gap is the near-total absence of trained AI/ML models operating on live checkpoint data to predict baggage delay, misrouting risk, or optimal rerouting before an incident occurs. The 2021 systematic review [7] confirms this at a field level, finding that predictive analytics in airports has so far been applied mainly to equipment maintenance rather than to the baggage flow itself, while the 2025 robotic-IoT study [8] explicitly calls for AI/ML to move baggage handling from reactive to proactive. This is the gap the architecture proposed in Section 3 is designed to close.

8. Future Research Directions

- Develop and benchmark delay-prediction models (e.g., gradient-boosted trees or sequence models) trained on historical checkpoint-timestamp and flight-schedule data, extending the predictive gap identified in Section 7.2.
- Fuse RFID/IoT scan data with machine-vision anomaly signals (building on Chen et al. [4]) into a single multimodal model to catch cases that neither data source alone would flag.
- Evaluate lightweight consortium-blockchain designs at full airport scale to determine whether the security gains shown by Si et al. [3] hold without unacceptable latency.
- Extend passenger-facing applications beyond status lookup to proactive ETA and misrouting-risk notifications, informed by the AI analytics engine.
- Pilot the proposed architecture at a single terminal with a phased rollout, following the incremental validation approach used in the Heathrow predictive-maintenance trial referenced in the systematic review [7].
- Investigate federated or privacy-preserving learning approaches so that airlines and airports can share training data for delay prediction without centralizing sensitive passenger information.

9. Conclusion

This literature survey reviewed eight studies spanning RFID, IoT, blockchain, and machine-vision approaches to airport baggage tracking, published between 2008 and 2025. The comparative analysis in Table 1 and Figure 3 shows that the field has matured considerably in real-time location tracking and, more recently, in data security through blockchain integration, but predictive capability remains the weakest and least explored dimension across nearly every reviewed system. The architecture proposed in Section 3, which layers an AI/ML analytics engine over an RFID/IoT sensing network, directly targets this gap and aligns with the objectives of reducing baggage misrouting, predicting delays, and improving passenger communication set out in Section 1.3. Future work, as outlined in Section 8, should focus on building and validating the predictive models themselves at real airport scale.

10. References

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